Markowitz portfolio

The aim of this lecture is an in-depth mathematical analysis of processes in which decisions have to be made and each choice involves risk.

1 Repetition

The investor consider N risky assets for which:

- *Rates of returns* of assets $R_1, R_2, \ldots, R_N$ are random variables with finite variances;
- *Expected rate of return* of i asset is $m_i = E(R_i)$;
- *Covariance between assets* i and j is $\sigma_{i,j} = Cov(R_i, R_j)$, and the covariance matrix is $\Sigma := [\sigma_{i,j}]_{i=1,\ldots,N, j=1,\ldots,N}$;

By *portfolio* means a vector $x = (x_1, x_2, \ldots, x_N)$ where each x_i means a share of portfolio in asset i.

- Negative value $x_i < 0$ means a *short position* in asset i, for example *short saling*, that means to borrow an asset i in amount x_i and for selling it immediately with obligation to buy back in the future;
- Negative value $x_i > 0$ means a *long position.*

We usually consider

- *Normalized portfolio,* that is

$$x_1 + x_2 + \ldots + x_N = 1;$$

- *Self financing portfolio,* that is

$$x_1 + x_2 + \ldots + x_N = 0.$$

Ingredients of portfolio. Let $x = (x_1, x_2, \ldots, x_N) = [x_1, x_2, \ldots, x_N]^T$ and $m := (m_1, m_2, \ldots, m_N) = [m_1, m_2, \ldots, m_N]^T$ (identify a vector with the corresponding single column matrix).

- *Expected rate of return of portfolio* x is

$$m_x := E\left(\sum_{i=1}^{N} x_i R_i\right) = \sum_{i=1}^{N} x_i m_i = x^T * m = \langle x, m\rangle;$$

- **Variance of portfolio** x is

$$\sigma_x^2 = Var\left(\sum_{i=1}^{N} x_i R_i\right) = \sum_{i=1}^{N}\sum_{j=1}^{N} x_i x_j \sigma_{i,j} = x^T * \Sigma * x,$$

  and we call σ_x-*volatility of portfolio* x.

1.1 Markowitz problem

For given $\tau \geq 0$, the investor finds *an optimal normalized portfolio* $x = (x_1, x_2, \ldots, x_N)$ such that solves the following **Markowitz problem**:

$$\max\left[\tau m_x - \sigma_x^2\right] \quad \text{such that} \quad \langle x, \mathbf{1}\rangle = 1,$$

where $\mathbf{1} = (1, 1, \ldots, 1)$.
Altenative formulations of Markowitz problem are:

$$\max\left[\tau \sum_{i=1}^{N} m_i x_i - \sum_{i=1}^{N}\sum_{j=1}^{N} x_i x_j \sigma_{i,j}\right] \quad \text{such that} \quad \sum_{i=1}^{N} x_i = 1.$$

or in matrix form:

$$\max[\tau\, m^T * x - x^T * \Sigma * x] \quad \text{such that} \quad \mathbf{1}^T * x = 1.$$

A portfolio x is *efficient* if there exists $\tau \geq 0$ for which x solves Morkowitz problem with τ.

1.2 Efficiency frontier

Let $x^*(\tau)$ be an efficient portfolio with τ.

- Then the plot $\left\{\left(\sigma^2_{x^*(\tau)}, m_{x^*(\tau)}\right) : \tau \geq 0\right\}$ is called an *efficiency frontier*.
- If Σ is positively definite, for every $\tau \geq 0$, every efficient portfolio $x^*(\tau)$ is unique and is *Pareto optimal*, that is there is no portfolio $\tilde{x}$ for which

$$m_{\tilde{x}} \geq m_{x^*(\tau)} \quad \text{and} \quad \sigma^2_{\tilde{x}} \leq \sigma^2_{x^*(\tau)}$$

  with at least one strict inequality;

- If $m_1 = m_2 = \ldots = m_N$, then every efficient portfolio is equal $x^*(0)$, also min variance portfolio, and then the efficiency frontier is *degenerated*.

1.3 Differentiation-notation and remarks

We write $x = (x_1, x_2, \ldots, x_N)$ as a column vector. Then linear functional in the form $x \in \mathbb{R}^N \mapsto \langle m, x\rangle = m^T * x$ is identified with the row wector $m^T = [m_1, m_2, \ldots, m_N]$. Because of that, for an open set $G \subset \mathbb{R}^N$, if $f : G \mapsto \mathbb{R}$ is differentiable, the gradient is a row vector

$$grad(f) = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_N}\right].$$

If f is twice continuously differentiable then the Hessian matrix

$$Hes(f) = \left[\frac{\partial f^2}{\partial x_i \partial x_j}\right]_{i=1,\ldots,N, j=1,\ldots,N}$$

is a symmetric matrix. If $Hes(f)$ is posively definite then f is strictly convex, and if is negatively definite then f is strictly concave.

2 Properties of efficient portfolio

For N assets for which the vector of expected rates of return is $m = (m_1, m_2, \ldots, m_N)$ and $\Sigma := [\sigma_{i,j}]_{i=1,2,\ldots,N, j=1,2,\ldots,N}$ is a covariance matrix, The optimal normalized portfolio for $\tau \geq 0$ solves

$$\max \tau m^T * x - x^T * \Sigma * x \quad \text{such that} \quad e^T * x = 1$$

where $e := (1, 1, \ldots, 1)$.

We recall the following assumption.

Assumption 1. *Assume that:*

(i) There is at least one pair (i, j) such that $m_i \neq m_j$;

(ii) The covariance matrix Σ is positively definite.

The first result is in order.

Proposition 1. *Define $q(x) = x^T * \Sigma * x$. Then $grad(q) = 2x^T\Sigma$ and $Hes(q) = 2\Sigma$.*

Proof. Observe that

$$q(x) := x^T * \Sigma * x = \sum_{i=1}^{N}\sum_{j=1}^{N} \sigma_{i,j} x_i x_j = \sum_{i=1}^{N} \sigma_{i,i} x_i^2 + \sum_{i=1}^{N}\sum_{j:j\neq i} \sigma_{i,j} x_i x_j.$$

Hence for every fixed i, we can write

$$\begin{aligned} q(x) &= \sigma_{i,i} x_i^2 + \sum_{j':j'\neq i} \sigma_{i,j'} x_i x_{j'} + \sum_{j':j'\neq i} \sigma_{j',i} x_{j'} x_i + \sum_{i',j':(i'\neq i)\wedge(j'\neq i)} \sigma_{i',j'} x_{i'} x_{j'} \\ &= \sigma_{i,i} x_i^2 + 2\sum_{j':j'\neq i} \sigma_{i,j'} x_i x_{j'} + \sum_{i',j':(i'\neq i)\wedge(j'\neq i)} \sigma_{i',j'} x_{i'} x_{j'}. \end{aligned}$$

As a result

$$\frac{\partial q}{\partial x_i} = 2\,\sigma_{i,i} x_i + 2\sum_{j:j\neq i} \sigma_{i,j} x_j = 2\sum_{j=1}^{N} \sigma_{i,j} x_j.$$

Thus,

$$grad(q) = \left[\frac{\partial q}{\partial x_1}, \frac{\partial q}{\partial x_2}, \ldots, \frac{\partial q}{\partial x_N}\right] = 2x^T * \Sigma.$$

As a results $\frac{\partial q}{\partial x_i \partial x_j} = 2\sigma_{i,j}$. Thus, the Hessian matrix of q is $Hes(q) = 2\Sigma$. □

The optimal normalized portfolio can be provided by the Lagrange multipliers methods:

$$\mathcal{L}_\tau(x, \lambda) = \tau m^T * x - \underbrace{x^T * \Sigma * x}_{=q(x)} + \lambda[e^T * x - 1].$$

Indeed, Applying Proposition 1 and Assumption 1 that Σ is positively definite we obtain that the function

$$f_\tau(s) := \tau m^T * x - x^T * \Sigma * x$$

is strictly concave since

$$Hes(f_\tau) = -2\Sigma.$$

is negatively definite. Hence for every $\tau \geq 0$, the Markowitz portfolio is unique. We denote this portfolio as $x^*(\tau)$. In particular, $x^*(0)$ - the min variance-portfolio is denoted by x^{MV}.

We now construct the formulas for $x^*(\tau)$ and in particular for x^{MV}. Observe that

$$grad(\mathcal{L}_\tau) = \tau\, m^T - 2\, x^T * \Sigma + \lambda\, e^T.$$

Thus, there is a unique solution of the Markowitz problem and satisfies the following system of equations:

$$\begin{cases} \tau\, m^T - 2\, x^T * \Sigma + \lambda\, e^T &= \mathbf{0} \\ e^T * x &= 1. \end{cases}$$

Rearranging the term and noting that $\Sigma^T = \Sigma$, we have

$$\begin{cases} \Sigma * x &= \frac{\tau}{2}\, m + \frac{\lambda}{2}\, e, \\ e^T * x &= 1. \end{cases}$$

Since Σ is positively definite we have

$$\begin{cases} x & = & \frac{\tau}{2}\,\Sigma^{-1} * m + \frac{\lambda}{2}\Sigma^{-1} * e \\ e^T * x & = & 1. \end{cases} \tag{1}$$

From the system (1) we have the following theorem.

Theorem 1. *Under Assumption 1, for each $\tau \geq 0$ a unique efficient portfolio $x^*(\tau)$ and in particular the min-variance portolio x^{MV} satisfy the following formulas:*

$$x^{MV} = \frac{\Sigma^{-1} * e}{e^T * \Sigma^{-1} * e} \tag{2}$$

$$x^*(\tau) = x^{MV} + \frac{\tau}{2} z^* \tag{3}$$

where

$$z^* = \Sigma^{-1} * m - \frac{e^T * \Sigma^{-1} * m}{e^T * \Sigma^{-1} * e}\Sigma^{-1} * e. \tag{4}$$

Moreover, $z^* \neq \mathbf{0}$ is a self-financing portfolio, that is $e^T * z^* = \mathbf{0}$.

Proof. We already know that the optimal Markowitz portfolio $x^*(\tau)$ is unique and satisfied (1). Multiplying both sides of the first equation from the left we have an equivalent system

$$\begin{cases} x & = & \frac{\tau}{2}\,\Sigma^{-1} * m + \frac{\lambda}{2}\Sigma^{-1} * e \\ e^T * x & = & \frac{\tau}{2}\,e^T * \Sigma^{-1} * m + \frac{\lambda}{2} e^T * \Sigma^{-1} * e \\ e^T * x & = & 1, \end{cases}$$

and hence from the third equation we get that

$$\begin{cases} x & = & \frac{\tau}{2}\,\Sigma^{-1} * m + \frac{\lambda}{2}\Sigma^{-1} * e \\ 1 & = & \frac{\tau}{2}\,e^T * \Sigma^{-1} * m + \frac{\lambda}{2} e^T * \Sigma^{-1} * e \end{cases}$$

Since Σ is positively definite, hence Σ^{-1} is positively definite, hence $e^T * \Sigma^{-1} * e > 0$. Thus,

$$\lambda = \frac{2 - \tau\, e^T * \Sigma^{-1} * m}{e^T * \Sigma^{-1} * e} := \lambda^*(\tau).$$

Hence

$$x^*(\tau) = \frac{\tau}{2}\, \Sigma^{-1} * m + \frac{\lambda^*(\tau)}{2} \Sigma^{-1} * e.$$

In particular, $\lambda^*(0) = \frac{2}{e^T * \Sigma^{-1} * e}$ and the mean-variance portfolio satisfies

$$x^*(0) = x^{MV} = \frac{\Sigma^{-1} * e}{e^T * \Sigma^{-1} * e}.$$

Hence we have derived the formula in (2).

Furthermore,

$$x^*(\tau) - x^{MV} = \frac{\tau}{2}\, \Sigma^{-1} * m + \frac{\lambda^*(\tau) - \lambda^*(0)}{2} \Sigma^{-1} * e$$

and

$$\lambda^*(\tau) - \lambda^*(0) = -\frac{\tau\, e^T * \Sigma^{-1} * m}{e^T * \Sigma^{-1} * e},$$

hence

$$x^*(\tau) - x^{MV} = \frac{\tau}{2}\, \Sigma^{-1} * m + \frac{\lambda^*(\tau) - \lambda^*(0)}{2} \Sigma^{-1} * e,$$

hence

$$x^*(\tau) - x^{MV} = \frac{\tau}{2} \left[\Sigma^{-1} * m - \frac{e^T * \Sigma^{-1} * m}{e^T * \Sigma^{-1} * e} \Sigma^{-1} * e \right] = \frac{\tau}{2} z^*.$$

Hence we have derived (3).

Now we show that z^* is self-financing. Indeed, multiplying both sides in (4) from the

left by e^T we immediately have

$$\begin{aligned} e^T * z^* &= e^T \left[\Sigma^{-1} * m - \underbrace{\frac{e^T * \Sigma^{-1} * m}{e^T * \Sigma^{-1} * e}}_{\text{a scalar}} (\Sigma^{-1} * e) \right] \\ &= e^T * \Sigma^{-1} * m - \underbrace{\frac{e^T * \Sigma^{-1} * m}{e^T * \Sigma^{-1} * e}}_{\text{a scalar}} (e^T * \Sigma^{-1} * e) \\ &= \mathbf{0}. \end{aligned}$$

□

Theorem 1 and particular (3) leads us two conclusion that the trajectory $\{x^*(\tau) : \tau \geq 0\}$ is a half-line in $\mathbb{R}^N$. We can provide the so called Two-Fund Theorem.

Theorem 2 (Two-Fund Theorem)**.** *For given $\tau_1 \geq 0$ and $\tau_2 \geq 0$, such that $\tau_1 \neq \tau_2$, every efficient porftfolio x^* can be written by*

$$x^* = \alpha x^*(\tau_1) + (1 - \alpha) x^*(\tau_2)$$

with some $\alpha \in \mathbb{R}$.

Proof. Let $\tau \geq 0$ be the parameter for which $x^* = x^*(\tau)$. From Theorem 1, we have the following formulas

$$\begin{aligned} x^*(\tau) &= x^{MV} + \frac{\tau}{2} z^*, \\ x^*(\tau_1) &= x^{MV} + \frac{\tau_1}{2} z^*, \\ x^*(\tau_2) &= x^{MV} + \frac{\tau_2}{2} z^*. \end{aligned}$$

for z^* defined in (4). Take α such that $\tau = \alpha\tau_1 + (1 - \alpha)\tau_2$. This is exactly $\alpha = \frac{\tau - \tau_2}{\tau_1 - \tau_2}$.

Then we obtain

$$\alpha x^*(\tau_1) + (1-\alpha)x^*(\tau_2) = x^{MV} + \frac{\alpha\tau_1 + (1-\alpha)\tau_2}{2} z^* = x^{MV} + \frac{\tau}{2} z^* = x^*(\tau) = x^*.$$

□

3 Properties of efficiency frontier

Let $m = (m_1, m_2, \ldots, m_N)$ be the vector of expected rates of assets $i = 1, 2, ..., N$, and $\Sigma = [\sigma_{i,j}]_{i=1,...,N, j=1,...,N}$ the covariance matrix or rates. From the previous lecture we know that the expected rate of return of portfolio x is

$$m_x = m^T * x$$

and the variance of portfolio is

$$\sigma_x^2 = x^T * \Sigma * x.$$

Let $x^*(\tau)$ be an efficient portfolio. Then from Theorem 1 it has the form

$$x^*(\tau) = x^{MV} + \frac{\tau}{2} z^*,$$

where recalling (4) we have

$$z^* = \underbrace{\Sigma^{-1} * m}_{\text{matrix } N\times 1} - \underbrace{\frac{e^T * \Sigma^{-1} * m}{e^T * \Sigma^{-1} * e}}_{\text{number}} \underbrace{\Sigma^{-1} * e}_{\text{matrix } N\times 1} . \tag{5}$$

Let us introduce the following numbers

$$\begin{cases} a &= m^T * \Sigma^{-1} * m > 0 \\ b &= e^T * \Sigma^{-1} * m \\ c &= e^T * \Sigma^{-1} * e > 0 \\ d &= ac - b^2 > 0. \end{cases} \tag{6}$$

For showing that $a > 0, c > 0$ and $d > 0$ we observe that the following formula $\langle x, y\rangle_{\Sigma^{-1}} = x^T * \Sigma^{-1} * y$ is an *inner product* in $\mathbb{R}^N$ and $||x||_{\Sigma^{-1}} = \sqrt{x^T * \Sigma^{-1} * x}$ is its norm. Since e and m are non-zero vectors, $a = ||m||^2_{\Sigma^{-1}} > 0$ and $c = ||e||^2_{\Sigma^{-1}} > 0$. Furthermore, we recognize $b = \langle e, m\rangle_{\Sigma^{-1}}$ hence

$$d = ||m||^2_{\Sigma^{-1}}||e||^2_{\Sigma^{-1}} - \langle e, m\rangle^2_{\Sigma^{-1}}.$$

The *Cauchy-Schwartz* inequlity yields $\langle e, m\rangle^2_{\Sigma^{-1}} \leq ||m||^2_{\Sigma^{-1}}||e||^2_{\Sigma^{-1}}$, hence $d \geq 0$. Assumption 1 that there are i, j such that $m_i \neq m_j$, hence the vectors e and m are *linearly independent*. Thus, the *Cauchy-Schwartz* inequlity yields the strict inequality $\langle e, m\rangle^2_{\Sigma^{-1}} < ||m||^2_{\Sigma^{-1}}||e||^2_{\Sigma^{-1}}$. Therefore, $d > 0$.
We are going to examine the shape of efficiency frontier $\{(\sigma^2_{x^*(\tau)}, m_{x^*(\tau)}) : \tau \geq 0\}$. Let us denote

$$\tilde{m}(\tau) := m_{x^*(\tau)} = m^T * x^*(\tau), \quad \text{and} \quad \tilde{\sigma}^2(\tau) := \sigma^2_{x^*(\tau)} = [x^*(\tau)]^T * \Sigma * x^*(\tau).$$

Theorem 3. *The efficiency frontier* $\{(\tilde{\sigma}^2(\tau), \tilde{m}(\tau)) : \tau \geq 0\}$ *is a parabola satisfying the following parametric plot*

$$\begin{cases} \tilde{m}(\tau) &= \frac{b}{c} + \frac{\tau}{2}\frac{d}{c}, \\ \tilde{\sigma}^2(\tau) &= \frac{1}{c} + \frac{\tau^2}{4}\frac{d}{c}, \quad \tau \geq 0, \end{cases}$$

where a, b, c *and* d *are defined in (6).*

Proof. From Theorem 1 and in particular (2) we get that

$$x^{MV} = \frac{1}{e^T * \Sigma^{-1} * e}\Sigma^{-1} * e = \frac{1}{c}\Sigma^{-1} * e.$$

Thus, from (5) can be express as follows

$$z^* = \Sigma^{-1} * m - \frac{b}{c}\Sigma^{-1} * e,$$

hence from Theorem 1 we have

$$x^*(\tau) = x^{MV} + \frac{\tau}{2} z^* = \frac{1}{c}\Sigma^{-1} * e + \frac{\tau}{2}\left(\Sigma^{-1} * m - \frac{b}{c}\Sigma^{-1} * e\right) = \frac{1 - \frac{\tau}{2}b}{c}\Sigma^{-1} * e + \frac{\tau}{2}\Sigma^{-1} * m. \tag{7}$$

Thus,

$$\begin{aligned} \tilde{m}(\tau) = m^T * x^*(\tau) &= \frac{1 - \frac{\tau}{2}b}{c} m^T * \Sigma^{-1} * e + \frac{\tau}{2} m^T * \Sigma^{-1} * m \\ &= \frac{b - \frac{\tau}{2}b^2}{c} + \frac{\tau}{2}a = \frac{b}{c} + \frac{\tau}{2}\frac{d}{c}. \end{aligned}$$

From (7) we conclude that the variance of portfolio is

$$\begin{aligned} \tilde{\sigma}^2(\tau) &:= [x^*(\tau)]^T * \Sigma * x^*(\tau) \\ &= \left[\frac{1 - \frac{\tau}{2}b}{c}\Sigma^{-1} * e + \frac{\tau}{2}\Sigma^{-1} * m\right]^T * \Sigma * \left[\frac{1 - \frac{\tau}{2}b}{c}\Sigma^{-1} * e + \frac{\tau}{2}\Sigma^{-1} * m\right] \\ &= \left[\frac{1 - \frac{\tau}{2}b}{c} e^T * \Sigma^{-1} + \frac{\tau}{2} m^T * \Sigma^{-1}\right] * \left[\frac{1 - \frac{\tau}{2}b}{c} * e + \frac{\tau}{2} * m\right] \\ &= \frac{\left(1 - \frac{\tau}{2}b\right)^2}{c} + \frac{(1 - \frac{\tau}{2}b)b}{c}\tau + \frac{\tau^2}{4}a = \frac{1}{c} + \frac{\tau^2}{4}\frac{d}{c}. \end{aligned}$$

□

Theorem 3 enables us to evaluate a direct dependence between $\tilde{m}(\tau)$ and $\tilde{\sigma}(\tau)$. From the (x, y) cartesian system with and $x = \tilde{\sigma}^2(\tau)$ and $y = \tilde{m}(\tau)$ we have the efficiency frontier at the following form

$$\begin{cases} x &= \frac{1}{c} + \frac{\tau^2}{4}\frac{d}{c}, \\ y &= \frac{b}{c} + \frac{\tau}{2}\frac{d}{c}, \quad \tau \geq 0, \end{cases}$$

Since $a > 0, c > 0$ and $d > 0$, hence $x \geq \frac{1}{c}$ and $y \geq \frac{b}{c}$. Moreover,

$$\frac{\tau}{2} = \frac{y - \frac{b}{c}}{\frac{d}{c}} = \frac{cy - b}{d},$$

hence the plot in the (x, y) system is

$$x = \frac{1}{c} + \left(\frac{c\,y - b}{d}\right)^2 \frac{d}{c} = \frac{1}{c} + \left(y - \frac{b}{c}\right)^2 \frac{c}{d},$$

hence

$$\left(y - \frac{b}{c}\right)^2 = 2\,\frac{d}{2c}\left(x - \frac{1}{c}\right),$$

is an upper arm of the parabola with a vertex at $\left(\frac{1}{c}, \frac{b}{c}\right)$. The focal distance is equal $\frac{d}{4c}$ and the focus $\left(\frac{4+d}{4c}, \frac{b}{c}\right)$ and the vertical directrix $x = \frac{4-d}{4c}$.

Furthermore, observe that for $\tau > 0$

$$\frac{d\tilde{\sigma}^2}{d\tau} = \frac{\tau}{2}\frac{d}{c} > 0, \quad \text{and} \quad \frac{d\tilde{m}}{d\tau} = \frac{d}{2c} > 0,$$

hence more risk tolerated investor selects a portfolio with higher expected rate of return and with higher variance.

References

[1] Evstigneev, I.V, Hens, T., Schenk-Hoppé, K.R. (2015). *Mathematical Financial Economics. A Basic Introduction*. Springer Cham Heidelberg New York Dordrecht London © Springer International Publishing Switzerland.